# PHY 130A
# Lecture Notes:
# Special Relativity and Relativistic Kinematics
# (Griffith's Chapter 3)

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# Chapter 3

# Relativistic Kinematics

## 3.1 Vectors and Change of Basis

One of the first concepts encountered in introductory physics is the vector as a geometric object with both a magnitude and a direction. In more advanced subjects, like quantum mechanics, we introduce the abstract definition of a vector space $V$ over a scalar field $F$ (e.g. the real or complex numbers). The vector space is closed under vector addition and scalar multiplication, and has the following defining properties:

**(1)** $v + w = v + w$ for all $v, w \in V$.
**(2)** $v + (w + x) = (v + w) + x$ for all $v, w, x \in V$.
**(3)** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
**(4)** For all $v \in V$ there exists $-v$ such that $v + (-v) = 0$.
**(5)** $c(v + w) = cv + cw$ for all $v, w \in V$ and $c \in F$.
**(6)** $1v = v$ for all $v \in V$ (note: $1 \in F$).
**(7)** $(c_1 + c_2)v = c_1 v + c_2 v$ for all $v \in V$ and $c_1, c_2 \in F$).
**(8)** $c_1(c + 2v) = (c_1 c_2)v$ for all $v \in V$ and $c_1, c_2 \in F$).

As you will show in the exercise below, additional basic properties are deducible from these axioms:

| | | |
|---|---|---|
| **(9)** | $0 \in V$ is unique | (Show $0' + v = v \implies 0' = 0$) |
| **(10)** | $0v = 0$ for all $v \in V$ | (Use $0 + 0 = 0$ for scalars) |
| **(11)** | $-v \in V$ is unique | (Show that $v + (-v') = 0 \implies -v = -v'$) |
| **(12)** | $(-1)v = -v$ for all $v \in V$ | (Use $1 + (-1) = 0$ for scalars) |

Note that we use the same symbol (0) for both the scalar zero and the zero vector; you must infer which one it refers to from context.

**Exercise 3.1:** (*) Deduce properties (9)-(12) from axioms (1)-(8) and known properties of the scalar field $F$. □

The vector space $\mathbb{R}^n$ was the inspiration for the abstract vector space, but there are many more examples: complex numbers as vectors over the field of complex numbers, $n \times m$ real matrices over the scalar field of real numbers with ordinary matrix addition and scalar multiplication, real-valued periodic functions $f(x)$ of a real variable with period $T$ over the scalar field of real numbers with ordinary addition and mulitiplication.

**Exercise 3.2:** The set $V$ is the rational numbers $\mathbb{R}$ and the field $F$ is the real numbers, with ordinary addition and multiplication. Show that $V$ is not a vector space.
**Exercise 3.3:** The set $V$ is the empty set. Is $V$ a vector space?□

**Exercise 3.4:** (*) For a vector space $V$ scalar multiplication is a null operation, that is $av = v$ for all $a \in F$ and $v \in V$. Deduce all of the vectors in the vector space $V$.

**Exercise 3.5:** (*) The set $V$ is the positive reals, with all reals as the scalar field $F$. For clarity in what follows, we'll use $\oplus$ for vector addition and $\odot$ for scalar multiplication. Vector addition is defined as multiplication of the real numbers, which we write as:

$$v \oplus w = v\, w \qquad v, w \in V$$

for $w, v \in V$. Scalar multiplication is exponentiation, which we write as:

$$c \odot v = v^c \qquad v \in V, c \in F$$

Show that $V$ is a vector space. What is the real number corresponding to the vector 0? □

**Exercise 3.6:** (*) For the previous exercise, verify explicitly that derived properties (10) and (12) hold for this vector space. □

A basis $\mathcal{B}$ for a vector space $V$ is an ordered set of vectors $\mathcal{B} = e_1, e_2, \cdots, e_m$ with the following properties:

- They are linearly independent, that is:

  $$c^1 e_1 + c^2 e_2 + \cdots + c^n e_n = 0$$

  only if $c^i = 0$ for all $i$.

- They span the space, that is for every $v \in V$ there exists a set of scalars $\{v^i \in F\}$ such that:

  $$v = \sum_i v^i e_i$$

  .

The scalar quantities $v^i$ are called the components of the vector, and they depend on the basis $\mathcal{B}$ chosen. This is crucial point to understand, so it bears repeating: **a vector $v$ exists independently of any basis, but its components $v^i$ depend on the basis chosen $\mathcal{B} = \{e_i\}$.**

**Exercise 3.7:** Show that the components $v^i$ of a vector $v$ for basis $\mathcal{B}$ are unique. (This is essential for many of the results that follow.) □

Throughout this course, we will use the Einstein summation convention, whereby there is an implicit sum over repeated dummy indices with one a superscript and the other a subscript, e.g.:

$$v = v^i e_i := \sum_i v^i e_i$$

Only **after** a basis $\mathcal{B}$ is chosen, it is possible to represent a vector as a column vector:

$$[v]_{\mathcal{B}} = \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}$$

or a row vector:

$$[v]_{\mathcal{B}}^T = \begin{pmatrix} v^1 & v^2 & \cdots & v^n \end{pmatrix}$$

.

**Exercise 3.8:** We can think of the complex numbers as 2-D vector space with the real numbers as scalars. Show that $e_1 = 1$ and $e_2 = i$ are a basis $\mathcal{B}$ for this vector space. For $v = a + bi$, write the column vector $[v]_{\mathcal{B}}$. This is why we are able to visualize the complex plane as a 2-D space. □

**Exercise 3.9:** (*) The Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

are a basis for the vector space of traceless $2 \times 2$ Hermitian matrices. Write the $2 \times 2$ matrix (an element of this vector space):

$$v = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}$$

as a column vector $[v]_{\mathcal{B}}$ with basis $\mathcal{B} = \{\sigma_x, \sigma_y, \sigma_z\}$ □

Suppose we have two bases $\mathcal{B}$ and $\mathcal{B}'$. We have:

$$\mathcal{B} = \{e_i\}$$

as before, but for the primed basis we will write:

$$\mathcal{B}' = \{e_{i'}\}.$$

Notice that the index $i$ is primed to indicate it is an index in the $\mathcal{B}'$ basis, and not the quantity $e$ itself. We'll justify this convention shortly.

Since each $e_i$ is a vector in $V$, and $\{e_{i'}\}$ is a basis, there must (by definition of basis, as above) exist scalar values $A^i_{j'}$ such that:

$$e_i = A_i^{j'} e_{j'}$$

(Remember there is an implicit sum across $j'$ here.) Similarly, since each $e_{i'}$ is a vector in $V$, and $\{e_i\}$ is a basis, there must be some scalar values $A^j_{i'}$ such that:

$$e_{i'} = A^j_{i'} e_j$$

Remember, the location of the prime matters, so $A^i_{j'}$ refers to something altogether different from $A^j_{i'}$. But they are related:

$$e_i = A_i^{j'} e_{j'} = A_i^{j'} A^k_{j'} e_k$$

which requires (recall that the components of a basis are unique) that:

$$A_i^{j'} A_{j'}^k = \delta_i^k$$

and similarly:

$$A_{j'}^i A_k^{j'} = \delta_k^i$$

Any vector $v$ can be expressed in terms of either basis $\mathcal{B}$ or basis $\mathcal{B}'$, which we write as:

$$v = v^i e_i = v^{j'} e_{j'}$$

but we also have:

$$e_{j'} = A_{j'}^i e_i \tag{3.1}$$

so that:

$$v^i e_i = v^{j'} e_{j'} = v^{j'} \left( A_{j'}^i \, e_i \right) = \left( A_{j'}^i \, v^{j'} \right) e_i$$

In a previous exercise you showed that the components of a vector within a basis (such as $\{e_i\}$ here) are unique. Therefore, we can conclude that for all $i$:

$$v^i = A_{j'}^i \, v^{j'}$$

In a similar fashion, we can find that:

$$v^{j'} = A_i^{j'} \, v^i$$

Compare this transformation rule to Equation 3.1, we see that the components $v^i$ transform via $A_i^{j'}$, whereas the basis vectors $e_i$ transform via $A_{j'}^i$. That is, they transform in a contary fashion. We call the vector components $v^i$ contravariant, and we indicate this with an upper index. We call the basis vector $e_i$ covariant, and we indicate this with the lower index.

| | index location | quantity | transformation | inverse transformation |
|---|---|---|---|---|
| $v^i$ | upper index | contravariant component | $v^{i'} = A_j^{i'} v^j$ | $v^i = A_{j'}^i v^{j'}$ |
| $e_i$ | lower index | covariant basis vector | $e_{i'} = A_{i'}^j e_j$ | $e_i = A_i^{j'} e_{j'}$ |

**Exercise 3.10:** Our first definition of a vector is a geometric object with a magnitude and direction. Suppose we scale each basis vector by a scalar $s$, to obtain a new basis. We must scale the components from the original basis by some scalar value in order to obtain new components for the new basis which preserve the magnitude of the vector. What is that scalar value? Does the "contra" in contravariant components make sense in this context? □

**Exercise 3.11:** Consider the previous exercise, but instead of scaling, we rotate the original basis vectors by an angle $\theta$ about some axis $w$ to obtain a new basis. Now we must rotate the components from the original basis about the axis $w$ by some angle, in order to obtain new components for the new basis which preserve the direction of the vector. What is that angle? Does the "contra" in contravariant components make sense in this context? □

## 3.2 Dual Vectors

For a vector space $V$ over a scalar field $F$, we define a dual vector $f$ as an $F$-valued linear function on $V$:

$$f : V \to F$$

That is, if $f$ is a dual vector, then for any $v \in V$ we have $f(v) \in F$. The linearity means that for any $v, w \in V$ and $c \in F$ we have also:

$$f(v + c\,w) = f(v) + c\,f(w)$$

The set of all dual vectors to a vector space $V$ is called the **dual space of** $V$, and it is denoted as $V^*$. As you will verify in the exercise below, the dual space is itself a vector space.

**Exercise 3.12:** For a vector space $V$ over scaler field $F$, show that its dual space ($V^*$) is itself a vector space. Vector addition (addition of dual vectors) and scalar multiplication are defined such that:

$$(f + cg)(v) = f(v) + cg(v) \qquad\qquad f, g \in V^*,\ v \in V,\ c \in F$$

□

Assume that $\mathcal{B} = \{e_i\}$ is a basis for the vector space $V$. The action of any dual vector $f \in V^*$ on any vector $v \in V$ is completely determined by its action of $f$ on the basis vectors $e_i$. For each i, we define a scalar quantity

$$f_i \coloneqq f(e_i)$$

and we see that:

$$f(v) = f(v^i e_i) = v^i f(e_i) = v^i f_i \tag{3.2}$$

which only depends on $f$ via the scalar quantities $f_i$.

For each $i$, we also define a dual vector $e^i \in V^*$ by:

$$e^i(v) = v^i$$

We say that $e^i$ picks off the $i$th component of $v$. If follows that

$$e^i(e_j) = \delta^i_j$$

where

$$\delta^i_j = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

because $e_j$ is a vector in $V$ whose components are all 0, except for the $j$th component, which is 1. You will show in the exercises below that the set of dual vectors $\{e^i\}$ are a basis for $V^*$, which we call the **dual basis** and denote as $\mathcal{B}^*$. You will also show that the scalar quanties $f_i$ are in fact the components of $f$ in the $\mathcal{B}$, so we have:

$$f = f_i e^i$$

. which shows a dual vector $f$ expanded in terms of its components $f_i$ and basis vectors $e^i$.

The dual basis $\mathcal{B}^*$ depends entirely on the basis $\mathcal{B}$ of $V$. As $\mathcal{B}^*$ has one basis vector defined for each basis vector in $\mathcal{B}$, we see that **if $V$ is finite dimensional, then it's dual space $V^*$ has the same dimension.**

**Exercise 3.13:** (*) Verify that $e^i(v + cw) = e^i(v) + ce^i(w)$ so that $e^i$ is indeed a dual vector.□

**Exercise 3.14:** Show that

$$f = f_i e^i$$

is a valid expansion for $f$, by verifying that

$$f(v) = v^i f_i$$

as required by Equation 3.2. This shows that $\{e^i\}$ spans the dual space.□

**Exercise 3.15:** Show that if: $a_i e^i = 0$ then, for all $i$ $a_i = 0$ This shows that $\{e^i\}$ is linearly independent. Hint: define the dual vector $a = a_i e^i$ and evaluate $a(e_i)$ for each $i$. □

**Exercise 3.16:** :(*) Consider the vector space $V = \mathbb{R}^3$ with the basis $e_1 = i, e_2 = j, e_3 = k$. For a vector $\vec{v} \in V$, write out the dual basis components $e^i(\vec{v})$ explicitly in terms of dot products. For a dual vector $f \in V^*$, write out it's components $f_i$. Write $f(\vec{v})$ expanded in terms of these components and basis vectors. □

We showed previously that the components of a vector $v$ tranform in a covariant fashion, so we call the $v^i$ the covariant components. Suppose that our basis transforms according to:

$$e_{i'} = A^j_{i'} e_j$$

then we see that:

$$f_{i'} = f(e_{i'}) = f(A^j_{i'} e_j) = A^j_{i'} f(e_j) = A^j_{i'} f_j$$

so the components of the dual vector $f_i$ transform covariantly. Notice how nicely our notation tracks covariance and contravariance. As long as we keep indices in the same posiiton on both side of equations, avoiding constructions like:

$$f^i = f(e_i) \qquad (wrong!)$$

then linearity ensures the covariance and contravariance of quantities can be read off from the location of the index. (This is an important practical matter, so take the time to understand it.)

To recap, we can expand a vector $v \in V$ in terms of any basis $\mathcal{B} = \{e_i\}$ and its scalar components $v^i \in F$:

$$v = v^i e_i.$$

We can likewise expand a dual vector $f \in V^*$ in terms of the dual basis $\mathcal{B}^*$ and its scalar components $f_i \in F$:

$$f = f^i e^i$$

Notice that the location of the indices in dual vectors is opposite that of vectors, reflecting that dual vectors have covariant components and vectors have contravariant components. We can think of the scalar components $v^i$ and $f_i$ in two complementary ways. First, $v^i$ and $f_i$ are the unique set of scalars which scale the basis vectors of $\mathcal{B}$ and $\mathcal{B}^*$ to the vectors $v$ and $f$. Secondly, the components can both be obtained by evaluating a dual vector:

$$v^i = e^i(v)$$

and

$$f_i = f(e_i)$$

In the vector space $\mathbb{R}^3$ we are familiar with obtaining the components of a vector by taking dot products. In a Hilbert space, we do so with an inner product. But here, we have not defined an inner product. We see that dual vectors are a generalization of the dot product idea.

**Exercise 3.17:** Using the definition, show that $e^i$ transforms contravariantly, as expected for a quantity with upper indices.

## 3.3 Introduction to Tensors

A traditional physics definition of a tensor is a generalization of scalars and vectors, to include quantities with muliple indices which transform in a specific way when changing the basis. This definition is perfectly reasonable, but it is not particularly insightful, and things (like the transformation rules) seem to come from nowhere. We are going to use a more modern definition of a tensor, which is equivalent, but clarifies the situation considerably.

Let $V$ be vector space and $V^*$ its dual space. We define an $(r, s)$ tensor as an $F$-valued multilinear function of $r$ vectors and $s$ dual vectors:

$$T : \underbrace{V \times V \times \cdots \times V}_{r \text{ times}} \times \underbrace{V^* \times V^* \times \cdots \times V^*}_{s \text{ times}} \to F$$

The scalar value returned by a tensor is written as:

$$T(v_1, v_2, \cdots, v_n, f_1, f_2, \cdots, f_s)$$

where $v_1, v_2, \cdots, v_n \in V$ and $f_1, f_2, \cdots, f_n \in V^*$ are **not** components of vectors or dual vectors, despite the indices. We define addition and scalar multiplication in the obvious manner, as clarified by example below. This makes the set of all $(r, s)$ tensors a vector space, which we call $\mathcal{T}^r_s(V)$.

The proliferation of indices makes talking about general tensors cumbersome, so let's work things out explicitly for a (1,1) tensor even though our conclusions apply to all tensors. We define tensor addition of two (1,1) tensors $T$ and $S$ by:

$$(T + S)(v, f) = T(v, f) + S(v, f)$$

with $v \in V$ and $f \in V^*$. We define scalar multiplication by:

$$(cT)(v, f) = cT(v, f)$$

with $c \in F$. Multilinearity means that:

$$\begin{aligned} T(v + cw, f) &= T(v, f) + c\,T(w, f) \\ T(v, f + cg) &= T(v, f) + c\,T(v, g) \end{aligned}$$

for $w \in V$ and $g \in V^*$.

Under the traditional definition, we consider tensors to be a generalization of the concept of scalars and vectors, so that scalar and vectors are examples of tensors. Is that the case under the new definition? The only immediately obvious connection is that a $(0, 1)$ tensor precisely reproduces the definition of a dual vector. It is also convenient to consider (0,0) tensors as scalars. That is, we can think of scalar as a scalar function with zero vector and zero dual vector arguments.

It is a bit more challenging to connect vectors to tensors under the new definition. Formally, we say that vectors are isomorphic to the (1,0) tensors by mapping each vector $v \in V$ to a (1,0) tensor $T_v$

$$v \mapsto T_v$$

defined by:

$$T_v(f) \coloneqq f(v).$$

Less formally, we are going to play a little game, you and I. You only have an ordinary vector $v$, but you are going to convince me that you have a (0,1) tensor. So I give you $f \in V^*$ and, quite cleverly really, you put your vector into my dual vector, and give me back $f(v)$. This shows me that you have an $F$-valued function of $V^*$, as required for a (0,1) tensor. But is it linear? I give $g \in V^*$ and you give me $g(v)$. I give you $f + cg$ with $c \in F$ and you give me:

$$(f + cg)(v) = f(v) + cg(v)$$

which verfies that you have a linear function. So you have now convinced me that you have a $(0, 1)$ tensor, even though you really only had a vector.

Next we'll consider a (2,0) tensor $T$. By definition $T$ as an $F$-valued multilinear function of two vectors:

$$T : V \times V \to F$$

and the scalar value returned for $v, w \in V$ is written as

$$T(v, w).$$

A tensor is completely described by its action on any set of basis vectors. The components of $T$ are defined as

$$T_{ij} \coloneqq T(e_i, e_j)$$

we can calculate

$$T(v, w) = T(v^i\, e_i, w^j\, e_j) = v^i T_{ij} w^j,$$

which shows we can calculate $T(v, w)$ from only the components $T_{ij}$.

We can clearly see the difference between a tensor and matrix. Remember that vectors are geometric objects that exist independently of any particular basis. Only after a basis has been selected, we can refer to them by their components, and arrange their components into row or column vectors. In exactly the same way, a tensor exists independently of any basis. But once we define a basis $\mathcal{B}$ we can refer to a vector by it's components. We can arrange the components of a vector into a row vector (denoted $[v]_\mathcal{B}$) or a column vector (denoted $[v]_\mathcal{B}^T$). Likewise, we can arrange the compments of a (2,0) tensor into a matrix, denoted $[T]_\mathcal{B}$. We can calculate the action of $T$ on vectors $v$ and $w$ via matrix multiplication:

$$T(v, w) = v^i\, T_{ij}\, w^j = [v]_\mathcal{B}^T\, [T]_\mathcal{B}\, [v]_\mathcal{B} = \begin{pmatrix} v^1\, v^2\, \cdots\, v^n \end{pmatrix} \begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22} & \cdots & T_{2n} \\ \vdots & \vdots & \ddots & \\ T_{n1} & T_{n2} & & T_{nn} \end{pmatrix} \begin{pmatrix} w^1 \\ w^2 \\ \vdots \\ w^n \end{pmatrix}$$

Another benefit of the modern definition is that we can derive the transformation law for the components of the tensor, instead of treating that as part of the definition.

**Exercise 3.18:** (*) Suppose that a basis $\mathcal{B}$ is related to another basis $\mathcal{B}'$ by:

$$e_{i'} = A^j_{i'} e_j$$

and

$$e_i = A^{j'}_i e_{j'}$$

Use multilinearity to show that:

$$T_{i'j'} = A^k_{i'} A^l_{j'} T_{kl}$$

and:

$$T_{ij} = A^{k'}_i A^{l'}_j T_{k'l'}$$

□

**Exercise 3.19:** For a vector space $V$ over scaler field $F$, show that the set of all (2,0) tensors is a vector space.

## 3.4 Non-degenerate Hermitian Forms and Metric Duals

In quantum mechanics, we saw that a Hilbert space $H$ is a vector space with an inner product defined. The inner product is positive definite:

$$\langle v|v\rangle > 0$$

for $v \in H$ and $v \neq 0$. It turns out that this requirement is too restrictive for our purposes (special relativity). We will need a slightly more general concept than the inner product. We will use the bar symbol to denote complex conjugation, so $\bar{z}$ denotes the complex conjugate of $z$ for $z \in \mathbb{C}$. If the bar symbol is applied to a real number, it has no effect.

A non-degenerate Hermitian form on a vector space $V$ is an $F$-valued function, denoted $(\cdot|\cdot)$, which assigns to any ordered pair of vectors $v$ and $w$ the scalar value denoted $(v|w)$, and has the following properties:

**(1)** $(v|w_1 + c\, w_2) = (v|w_1) + c\,(v|w_2)$ for all $v, w_1, w_2 \in V$ and $c \in F$.
**(2)** $(v|w) = \overline{(w|v)}$ for all $v, w \in V$.
**(3)** For $v \in V$ and $v \neq 0$ there exists $w \in V$ with $(v|w) \neq 0$.

Now hopefully properties (1) and (2) are quite familiar properties of the inner products you already encountered in quantum mechanics, but property (3) is new. Notice, however, that property (3) is automatically satisfied for a Hilbert space, by choosing $w = v$ and appealing to positive definiteness. So conceptually, a non-degenerate form is an inner product but for the requirement of positive definiteness, which is loosened to property (3) instead.

**Exercise 3.20:** Show that for any non-degenerate Hermitian form $(\cdot|\cdot)$:

$$(v_1 + c\, v_2|w) = (v_1|w) + \bar{c}(v_2|w)$$

for all $v_1, v_2, w \in V$ and $c \in F$ $\square$

Given a vector $v \in V$ with non-degenerate Hermitian form $(\cdot|\cdot)$ we can define a dual vector $\tilde{v}$ by:

$$\tilde{v}(w) = (v|w)$$

for all $w \in V$. We call the dual vector $\tilde{v}$ the metric dual of $v$. We sometimes write:

$$\widetilde{v} = (v|\cdot)$$

.

Now consider the map $L : V \to V^*$ that sends a vector to it's metric dual:

$$L(v) \coloneqq \tilde{v}$$

**Exercise 3.21:** Show that $L$ is conjugate linear, that is for $v_1, v_2 \in V$ and $c \in F$ then

$$L(cv_1 + v_2) = c^* L(v_1) + L(v_2)$$

Hint: $(L(cv_1 + v_2))(w) = (c\, v_1 + v_2|w)$ $\square$

**Exercise 3.22:** Show that $L$ sends zero to zero. That is, show $L(0) = 0$. $\square$

**Exercise 3.23:** Show that $L$ is a one-to-one map of $V \to V^*$, that is, show that:

$$L(w) = L(v) \implies w = v$$

Hint: use non-degeneracy, i.e. if for all $x \in V$

$$(v|x) = 0$$

then

$$v = 0$$

$\square$

**Exercise 3.24:** Show that the only vector $L$ sends to zero is zero, that is:

$$L(w) = 0 \implies w = 0$$

$\square$

For a finite-dimensional vector space $V$ recall that the dual space $V^*$ has the same dimension as $V$. Let $\mathcal{B} = \{e_i\}$ be a basis for $V$. Using the results of the previous exercises, we can show that the set of dual vectors $\{L(e_i)\}$ is a basis for $V^*$. Because $L$ is one-to-one, $\{L(e_i)\}$ has the same number of elements as $\mathcal{B}$. As long as they are linearly indepdent, they will span $V^*$, because $V$ and $V^*$ have the same dimension.

To prove that $\{L(e_i)\}$ is linearly independent, assume there is some set of scalars $a^i$ such that:

$$a^i L(e_i) = 0$$

Then define a vector $v \in V$ by

$$v := a^i e_i$$

We have:

$$L(v) = L(a^i e_i) = a^i L(e_i) = 0$$

This implies (see previous exercises) that $v = 0$, and so:

$$a^i = 0$$

which shows that $\{L(e_i)\}$ are linearly independent. Therefore $\{L(e_i)\}$ is a basis for $V^*$

**Exercise 3.25:** Given that $\{L(e_i)\}$ is a basis for $V^*$, show that the mapping $L$ from $V$ to $V^*$ is onto. That is, show that for any $b \in V^*$, there is an $a \in V$ such that

$$L(a) = b$$

. □

By now we have shown that the mapping $L$ is both one-to-one and onto, which means that it is an invertible mapping. We can move freely from a vector to its metric dual and back again with no ambiguity.

For a Hilbert space $H$ with inner product $\langle \cdot | \cdot \rangle$. In Dirac notation, a vector $\psi \in H$ is written as a ket $|\psi\rangle$. It's metric dual $L(\psi)$ is written as bra $\langle \psi|$.

**Exercise 3.26:** (*) For $\psi \in H$ and $c \in F$, show that

$$\langle c\psi| = \overline{c}\, \langle \psi|$$

□

That completes the math prelimineraries we will need for our invariance-based treatment of special relativity.